Journal of Cleaner Production 24 (2012) 30–35

Contents lists available at SciVerse ScienceDirect

Journal of Cleaner Production

journal homepage: www.elsevier.com/locate/jclepro

Comparison of methodologies for estimating the carbon footprint – case study of office paper

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ARTICLE INFO

Article history:
Received 9 August 2011
Received in revised form
4 November 2011
Accepted 7 November 2011
Available online 16 November 2011

Keywords:
CEPI framework
Global warming potential
ISO 14040/14044
Office paper
PAS 2050
Wood product

ABSTRACT

Currently there are several methodologies available for estimating the carbon footprint of products. In this study a comparison has been made between the outcomes and the implications of three different methodologies applied to office paper: (1) the ISO 14040/14044 standards limited to the analysis of greenhouse gas (GHG) emissions and the corresponding impact category global warming; (2) the PAS 2050; and (3) the Confederation of European Paper Industries (CEPI) framework. The carbon footprint of office paper from cradle-to-costumer is 4.64, 4.74 and 4.29 g CO_2eq per A4 sheet according to, respectively, the ISO 14040/14044 standards, the PAS 2050 and the CEPI framework. The ISO 14040/14044 standard methodology allows the quantification of 98% of the total GHG emissions with the smallest effort in data collection. The major hot spots are the stages of eucalypt pulp and office paper production and chemical and fuel production for all methodologies. General methodologies such as those analysed in this study are not enough for the comparison of products. More specific rules, such as Product Category Rules, that limit the degree of freedom in the choice of the functional unit, system boundary, allocation rules, data quality, between others, should be developed.

1. Introduction

Nowadays there is widespread recognition that global warming is a major threat to the environment and economic development of the world. Rising levels of greenhouse gases (GHGs) in the atmosphere, mainly resulting from human activities, have enhanced the greenhouse effect and, consequently, caused a rise on the temperature of the Earth (IPCC, 2007). As a response to the threat of climate change, the international community has been developing and implementing numerous initiatives to decrease the GHG concentrations in the atmosphere. Such initiatives rely on the assessment, monitoring, reporting and verification of GHG emissions and removals to improve the scientific knowledge and provide supporting information for the establishment of climate policies.

Life cycle thinking tools, in particular carbon footprint, are becoming popular to assess GHG emissions of products (Guinée et al., 2011; Pandey et al., 2011). The carbon footprint of a product is usually defined as a quantification of the GHG emissions during the life cycle of the product and is being increasingly applied for multiple purposes, for example: to communicate the carbon footprint to customers, to facilitate the development and implementation of GHG management strategies across product life cycles, to identify potential opportunities for GHG mitigation along the supply chain, to track performance and progress in reducing GHG emissions over time and to assist the consumer in the selection of products with the least climate change impact.

Carbon footprint has been driven by nongovernmental organisations, companies and various private initiatives rather than by the scientific community (Weidema et al., 2008). This resulted in many methodological approaches for calculating the carbon footprint. The lack of a harmonised methodology can lead to misleading results due to the use of methodologies that favour the results. This happens mainly when carbon footprint is applied for marketing purposes. In addition, the lack of a harmonised methodology can hinder or make difficult comparisons between products. The development of harmonised, transparent and accurate methodologies is therefore of high importance.

In this context, the British Standards Institution (BSI) published in 2008 the Publicly Available Specification (PAS) 2050 which specifies the requirements to assess the life cycle GHG emissions of goods and services (BSI, 2008). The International Organization for Standardization (ISO) is working on a standard that will detail the principles and framework requirements for the quantification and communication of the carbon footprint of products (ISO 14067). The work of ISO on this standard started in 2008 and is planned to

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doi:10.1016/j.jclepro.2011.11.005

be completed in 2012. The Greenhouse Gas Protocol Initiative convened by the World Resources Institute (WRI) and the World Business Council for Sustainable Development (WBCSD) developed a standard to quantify and report the GHG emissions throughout the life cycle of a product (WRI/WBCSD, 2011). These three methodologies have been built from the existing Life Cycle Assessment (LCA) methodology established through ISO 14040 and 14044 standards (ISO, 2006a,b). However, these contain further provisions to a uniform quantification of GHG emissions in particular. Methodological approaches for carbon footprint calculation are also being developed by some industrial sectors. In 2007, the Confederation of European Paper Industries (CEPI) published a common framework for the calculation of the carbon footprint of paper and board products (hereafter referred to as CEPI framework) (CEPI, 2007). The pulp and paper sector is highly committed with determining the carbon footprint of its products and a proof of this is the publication of the CEPI framework as well as some studies presenting the carbon footprint of specific products of this sector (e.g., Boguski, 2010; Carbon Trust, 2006; Eriksson et al., 2010; Nors et al., 2009; Pihkola et al., 2010).

In this study a comparison is made between the outcomes and implications of adopting three different methodological approaches to determine the carbon footprint: (1) the ISO 14040/14044 standards, limited to the analysis of GHG emissions and the corresponding impact category global warming; (2) the PAS 2050; and (3) the CEPI framework. The office paper has been selected as case study to illustrate this comparison. The analysed office paper is used for printing and writing and is produced from virgin fibre chemical pulp. About 95% of this pulp consists of eucalypt bleached Kraft pulp that is directly integrated in the office paper production process and about 5% is pine bleached Kraft pulp received in bales.

2. Methods

2.1. Functional unit

The functional unit is the production of one A4 sheet of office paper with a standard weight of 80 g/m^2. The methodologies under analysis in this study provide similar guidance for the establishment of the functional unit, but none of them suggest functional units for particular products.

2.2. System definition and boundary

2.2.1. Stages included

The ISO 14040/14044 standards refer that the system boundary should be consistent with the goal of the study. These standards allow the consideration of the full life cycle (cradle-to-grave) or only parts of the life cycle. The PAS 2050 proposes cradle-to-grave or cradle-to-gate assessments. On the other hand, the CEPI framework recommends performing a cradle-to-gate or a cradle-to-customer study. This framework proposes the exclusion of end-of-life emissions due to the associated high uncertainty, the potential for distorted comparisons between footprints, and the almost complete lack of control of these emissions by the product manufacturer.

In this study, the carbon footprint has been determined from cradle-to-customer, i.e. up to the arrival at the distribution platform (the first business transaction), which is consistent with the three methodological approaches. Delivery to the final consumer is not considered due to the lack of accurate data on the distances travelled. Furthermore, the paper end-of-life is excluded from the boundary because the associated emissions are very uncertain as the alternatives for the final disposal are not known for every country.

As shown in Fig. 1, the following stages are generically considered.

1. *Eucalypt wood production*. Consists of cradle-to-gate production of eucalypt wood, including the entire operations carried out in site preparation (stump chipping, clearing and soil scarification), stand establishment (planting), stand tending (cleaning, fertilising, soil loosening and selection of coppice stems), logging (harvesting and forwarding) and infrastructure establishment (roads and firebreaks). It also includes the production of fertilisers and fuels used in the forest operations.
2. *Pine pulp production*. Consists of cradle-to-gate production of pine pulp and includes the production of elemental chlorine free (ECF) bleached Kraft pulp delivered in bales, the production and transport of raw and ancillary materials (pine wood and chemicals) and energy (electricity and fuels).
3. *Eucalypt pulp and office paper production*. Consists of gate-to-gate integrated production of eucalypt pulp and office paper and comprises wood debarking and chipping, wood cooking (with sodium hydroxide and sodium sulphide), brownstock washing and screening, ECF pulp bleaching using chlorine dioxide produced on site, pulping of pine pulp bales, pulp refining, cleaning and screening, broke recovery, paper machine, paper finishing, chlorine dioxide production, chemical recovery (to recover the cooking chemicals while producing energy from black liquor), energy generation from bark, black liquor and natural gas, and wastewater treatment in an activated sludge plant.
4. *Chemical and fuel production*. Consists of cradle-to-gate production of chemicals and fuels to be consumed in eucalypt pulp and office paper production.
5. *Solid waste end-of-life*. Consists of the gate-to-gate end-of-life of industrial waste generated in eucalypt pulp and office paper production. Inorganic waste such as grits, dregs and lime sludge are disposed in an industrial landfill whereas organic waste, such as wood yard debris and wastewater treatment sludge, are mainly applied in agricultural and forest soils.
6. *Avoided electricity production*. Consists of cradle-to-gate production of avoided electricity in the national grid (only used in the ISO 14040/14044 and PAS 2050 methodologies to avoid allocation).
7. *Transports*. Consists of transport of the materials consumed in eucalypt pulp and office paper production (eucalypt wood, pine pulp, chemicals and fuels), transport of wastes produced in the eucalypt pulp and office paper production stage and transport of the office paper to the distribution platform.

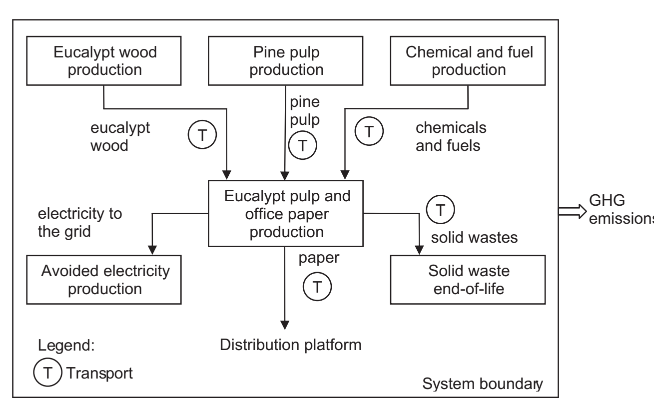

Fig. 1. Stages considered within the system boundary of the carbon footprint of office paper.

2.2.2. Cut-off criteria

Cut-off criteria have been established in order to decide which processes should be included within the boundaries. The ISO 14040/14044 standards do not suggest quantified thresholds but refer that the cut-off criteria should be based on mass, energy, and environmental significance. In this study, under this methodology, the materials that represent less than 1% of the mass of the produced paper have been left outside the boundaries. As significant emissions should not be left outside the boundaries, the production of hydrogen peroxide and sodium chlorate used in the pulp production processes have been included within the boundaries, regardless of the mass-based cut-off criterion, because they are energy-intensive processes (Strömberg et al., 1997).

The PAS 2050 recommends the inclusion of all sources of emissions that contribute to more than 1% of the anticipated life cycle GHG emissions of the functional unit and, in addition, to include at least 95% of the anticipated life cycle GHG emissions of the functional unit. Besides, where less than 100% of the anticipated life cycle GHG emissions have been determined, the assessed emissions should be scaled to represent 100% of the GHG emissions associated with the functional unit. In this study, under this methodology, all the processes generating GHG emissions have been included inside the boundary in order to comply with the 100% criterion.

The CEPI framework defends the inclusion of 90% of all GHG emissions within the boundaries which has been adopted in this study.

The adoption of different cut-off criteria for the three approaches resulted in the inclusion of a different number of chemicals, fuels and industrial wastes within the system boundaries as shown in Table 1.

2.2.3. Other exclusions

The production of capital goods (buildings, machinery and equipment), the transport of consumers to and from the point of purchase and the transport of employees to and from their place of work have also been excluded from the system boundary. This is in compliance with PAS 2050. Both the ISO 14040/14044 standards and the CEPI framework do not provide any specific procedure for these issues.

2.3. GHGs and global warming potentials

The GHGs considered in this study, whenever they were available, are those for which the last assessment report of the Intergovernmental Panel on Climate Change (IPCC, 2007) assigns a global warming potential (GWP), as recommended by PAS 2050. The GWPs adopted in this study are also from IPCC (2007) for a timeframe of 100 years, as stated by PAS 2050. Neither the ISO 14040/14044 standards nor the CEPI framework provide specific guidance on which GHGs and respective GWPs should be used.

2.4. Biomass-based carbon

Biomass-based carbon dioxide (CO_2) emissions have been considered to be neutral as they are balanced by CO_2 sequestration in forest, providing that forest is sustainably managed and that there are no changes on land use, which is the case of the forests that provide wood for the office paper manufacturing. Therefore, these emissions have not been included in this study. This is a common simplification in both LCA and carbon footprint studies (e.g., Dias et al., 2007a; González-García et al., 2010; Ross and Evans, 2002) and is recommended by the PAS 2050 and the CEPI framework. However, this issue is not focused by ISO 14040/14044 standards.

Table 1
Number of materials considered within the system boundary of the carbon footprint of office paper.

Material	Number of materials considered within the system boundary		
	ISO 14040/14044	PAS 2050	CEPI framework
Chemicals	10	20	3
Fuels	2	5	1
Industrial wastes	6	8	0

Biomass-based carbon storage in paper, both in use and in solid waste disposal sites, has not been considered since the paper use and end-of-life stages have been excluded. It should be noted that the PAS 2050 states that the carbon storage in products should be deducted from the total GHG emissions arising from the product life cycle. In addition, PAS 2050 provides a calculation procedure applicable to eligible products, i.e. non-food products, except those with a short life span for which the carbon storage is induced by human intervention. The CEPI framework also focuses the carbon storage in forest products and discusses several approaches for determining the net carbon sequestration in the forest sector but does not propose a specific calculation procedure. The ISO 14040/14044 standards disregard carbon storage in products.

2.5. Data collection

The ISO 14040/14044 standards provide robust guidance concerning data collection procedures for LCA studies that are also appropriate in the context of the carbon footprint calculation. The PAS 2050 provides similar general procedures and also more specific procedures for GHG data collection. According to this methodology, primary activity data (quantitative direct measurement of an activity that, when multiplied by an emission factor, determines the GHG emissions) should be collected for the processes owned, operated or controlled by the organisation implementing the PAS and contributing 10% or more to the upstream GHG emissions. For the other processes, the PAS 2050 allows the use of secondary data (derived from sources other than direct measurement such as peer-reviewed literature or databases). The CEPI framework does not focus this issue in particular. In this study, the data collected for each process are the same for all the methodologies and consist of inputs to the processes (materials and energy) as well as outputs from the processes (GHG emissions, products, co-products, energy and waste). Following PAS 2050 recommendation, primary activity data have been collected for the processes controlled by the paper producing company. Whenever possible, primary activity data have also been collected for other processes. The data sources used in this study, both for activity data and emission factors, are summarised in Table 2.

Fuel consumption (diesel oil and gasoline) by the eucalypt forest management activities have been estimated based on the type and number of forest operations, the effective work-time needed to perform each operation and the corresponding fuel consumption per hour of machine work, as described in Dias et al. (2007b). These primary activity data have been provided by a leading eucalypt woodland company. GHG emissions deriving from fuel combustion, which include fossil CO_2, methane (CH_4) and nitrous oxide (N_2O), have been calculated using emission factors from IPCC (2006). Data on fertilisers usage in eucalypt wood production are also primary activity data provided by a leading eucalypt woodland company. Data on fuel and fertiliser production are from the Ecoinvent database (Ecoinvent, 2010).

Data on raw and ancillary materials, fuels, electricity and wastes consumed/generated in the integrated production of eucalypt pulp

Table 2
Data sources for the activity data and emission factors used to estimate the carbon footprint of office paper.

Stage	Data source	
	Activity data	Emission factor
Eucalypt wood production		
Fuel and fertiliser use	Eucalypt woodland company	–
GHG emissions from fuel burning	–	IPCC (2006)
GHG emissions from fuel and fertiliser production	–	Ecoinvent (2010)
Pine pulp production	–	Ecoinvent (2010)
Eucalypt pulp and office paper production	Paper company	IPCC (2006)
Chemical and fuel production		
PCC	PCC company	Ecoinvent (2010) [a]
Optical brightener	–	KCL (1999)
Other chemicals and fuels	–	Ecoinvent (2010)
Solid waste end-of-life	Paper company	–
Avoided electricity production	–	Ecoinvent (2010)
Transports		
Distances	Paper company	–
Fuel use	Volvo (2006)	–
GHG emissions from fuel burning and production	–	Ecoinvent (2010)

[a] As the PCC production process has no on-site GHG emissions, the emission factors sourced from Ecoinvent (2010) are for the production of the raw materials and electricity used in PCC production.

and office paper have been provided by the paper producing company and are average data from the year 2006. The GHGs considered in the eucalypt pulp and office paper production process (fossil CO_2, CH_4 and N_2O) from fossil fuel combustion have been estimated considering the quantity of fuels consumed and specific emission factors for each fuel (IPCC, 2006). Fossil fuels are used for energy production, internal transports and also in lime kilns (included in the chemical recovery cycle). CO_2 emissions from the burning of make-up calcium carbonate in lime kilns have also been included.

Data concerning the production of pine pulp, chemicals, fuels and electricity in the grid are from the Ecoinvent database (Ecoinvent, 2010) and comprise several GHGs. One exception is the production of optical brightener, for which data are from KCL-EcoData (KCL, 1999). Another exception is the production of precipitated calcium carbonate (PCC), for which primary activity data have been directly obtained from the producer and data concerning the production of the raw materials and electricity used in its production are from Ecoinvent (2010).

The transport profiles (distances travelled, mean of transport and load state in the return journey) for the materials and wastes transported to or from the paper mill have been provided by the paper producing company. Data on diesel consumption by trucks are from Volvo (2006), as well as truck load capacity. GHG emission factors for transports are from the Ecoinvent database (Ecoinvent, 2010).

The end-of-life options for the industrial wastes produced in the eucalypt pulp and office paper mill depend on the waste properties. Inorganic wastes are disposed in an industrial landfill where decay, and consequently GHG emissions, has been assumed to be insignificant. Organic wastes are mainly applied into agricultural and forest soils where decay occurs in aerobic conditions leading to biomass-based CO_2 emissions, which are considered to be neutral.

2.6. Multifunctionality and allocation

In multifunctional processes, it is necessary to allocate the GHG emissions of these processes to the product under analysis. In this study, the eucalypt pulp and office paper production is a multiple output process that produces both paper and electricity that is sold to the national grid. This electricity is produced in a combined heat and power (CHP) system fuelled by natural gas.

The ISO 14044 standard presents a hierarchy of procedures to solve the allocation problem in LCA studies that can also be applied to carbon footprint studies. Wherever possible, allocation should be avoided by unit process division or by system boundary expansion. Where allocation cannot be avoided, the GHG emissions of the process should be partitioned based on physical relationships. Where physical relationships cannot be used, the allocation should be done using other criteria such as the economic value of the products. In this study, under this methodology, the allocation of GHG emissions between paper and electricity has been avoided through the expansion of the system boundary (avoided burdens) assuming that the sold electricity avoids the production of an equivalent amount of electricity in the national grid.

The approach of PAS 2050 to deal with allocation also relies on a hierarchy. Firstly, allocation should be avoided by unit process division or by system boundary expansion as in ISO 14044. Where allocation avoidance is not practicable, the GHG emissions should be allocated in accordance with the economic value of the products. The PAS 2050 also refers that the avoided emissions resulting from the co-production of electricity should be based on the average GHG emissions intensity of grid electricity. This recommendation has been adopted to avoid the allocation of GHG emissions between paper and electricity and, as in the ISO 14040/14044 methodology, the exportation of electricity to the grid was considered to avoid the production of the same amount of electricity in the grid.

The CEPI framework suggests the application of the ISO 14044 rules. However, for the particular case of allocation of emissions from CHP systems between electricity and steam, it refers to the CHP allocation options available on NCASI (2005), which are based on the amount of fuel used to produce each energy output (efficiency method) or on the amount of useful energy in each energy output (heat content and work potential methods). The heat content method has been applied in this study based on primary activity data concerning the amount of electricity produced, the mass and specific enthalpy of the steam produced and the mass and specific enthalpy of the returned condensates. Emission factors of 0.079 kg CO_2eq per MJ of electricity and 0.079 kg CO_2eq per MJ of steam (low-pressure steam minus returned condensates) have been obtained. Therefore, under the CEPI framework the GHG emissions associated with the exported electricity are not attributed to the production of eucalypt pulp and office paper and consequently the allocation of GHG emissions between paper and electricity is not necessary.

3. Results

The obtained carbon footprint of office paper has been 4.64, 4.74 and 4.29 g CO_2eq per A4 sheet, respectively, with the ISO 14040/14044 standards, the PAS 2050 and the CEPI framework. The result obtained with the ISO 14040/14044 standards represents 98% of the result obtained with the PAS 2050 which considers the totality of the GHG emissions. These two methodologies only differ in the cut-off rule. The methodology in accordance with the ISO 14040/14044 standards takes into account almost the totality of the emissions with a much smaller effort in data collection. The CEPI framework gave the lowest carbon footprint because only 90% of the GHG emissions have been accounted for. This cut-off rule lead to lower GHG emissions by the stage of chemical and fuel production because only the production of PCC, sodium chlorate, starch (surface) and natural gas have been considered. The GHG emissions from the transport stage are also smaller because only the transport of office paper to the distribution platform has been considered.

Fig. 2 shows the relative contribution of each stage of the life cycle of office paper to the total carbon footprint. The major hot spots for all the methodologies are the stages of eucalypt pulp and office paper production and chemical and fuel production. The eucalypt pulp and office paper production stage has the largest contribution for both the ISO 14040/14044 standards and the CEPI framework, accounting, respectively, for 44 and 46% of the total carbon footprint. For the PAS 2050, the stages of eucalypt pulp and office paper production and chemical and fuel production have similar contributions of 43% each, as, in this case, all the chemicals and fuels have been included in the assessment. The remaining stages have individual contributions smaller than 10%.

The main source of GHG emissions in the eucalypt pulp and office paper production stage is the burning of natural gas to produce energy, representing almost 80% of the total GHG emissions of this stage. The largest source of GHG emissions in the stage of chemical and fuel production, representing about 45–54% of the total carbon footprint of this stage, is the production of PCC which is the main raw material of office paper besides fibres. Other important contributions to the total carbon footprint of this stage are the production of natural gas (16–19%), the production of sodium chlorate (13–15%) and the production of surface starch (11–12%).

The adoption of different allocation procedures led to different results for the eucalypt pulp and office paper production. The carbon footprint of this stage is 2.05 g CO_2eq per A4 sheet using both the ISO 14040/14044 standards and the PAS 2050 and 1.96 g CO_2eq per A4 sheet using the CEPI framework. The net carbon footprint of this stage, obtained by subtracting the GHG emissions from the avoided electricity, is 1.91 g CO_2eq per A4 sheet with the ISO 14040/14044 standards and the PAS 2050.

4. Discussion

The methodologies applied in this study gave different results for the carbon footprint of office paper as they rely on different cut-off criteria and allocation procedures.

The cut-off criteria adopted when using the ISO 14040/14044 standards proved to be adequate because it allows the quantification of 98% of the GHG emissions with a relatively small effort in data collection. The methodology compliant with the PAS 2050 addresses 100% of the GHG emissions with a much larger effort for data collection (Table 1). The application of the cut-off rule recommended by the CEPI framework also requires great effort in data collection for the quantification of 100% of the GHG emissions in a first step, although only 90% of the GHG emissions are reported. In practice, the quantification of 100% of the GHG emissions may be unfeasible due to the lack of data for some processes.

The methodologies compliant with the ISO 14040/14044 standards and the PAS 2050 avoid the allocation of the GHG emissions between paper and electricity in the stage of eucalypt pulp and office paper production by system boundary expansion (avoided burdens). This approach is often adopted in pulp and paper studies (Dias et al., 2002; Vieira et al., 2010). Under the CEPI framework there is no need of allocation between paper and electricity because it is done between electricity and steam produced in the CHP system and, therefore, the emissions from the exported electricity are not attributed to the integrated eucalypt pulp and office paper production. The avoided burdens approach generates a smaller net carbon footprint (GHG emissions from this stage minus avoided GHG emissions) than the CEPI framework because the GHG emission factor for the production of electricity in the grid is larger than that for the electricity produced in the CHP system from natural gas (600 g CO_2eq/kWh and 285 g CO_2eq/kWh, respectively). The allocation procedure suggested by the CEPI framework is more favourable for countries where the GHG emission factor for the production of electricity in the grid is lower than that for the electricity produced in the CHP system. This is the case of countries where the grid electricity mix is mostly based on renewable or nuclear energy.

The choice of the functional unit requires special attention in order to allow comparisons between products without bias. In LCA or carbon footprint studies conducted for office paper it is usual to consider one tonne of paper as functional unit (Counsell and Allwood, 2007; Dias et al., 2002; Dias et al., 2007a; Lopes et al., 2003; Vieira et al., 2010). However, when carbon footprint is used for labelling purposes, consumers could be more able to understand its meaning if it is expressed per ream or per sheet. These functional units could also be more appropriate for the comparison of the carbon footprint of two types of office paper with different standard weight (e.g., 80 g/m^2 versus 90 g/m^2), because the function delivered by one tonne of each paper is different.

The results obtained in this study, expressed per tonne of office paper, are 930, 950 and 860 kg CO_2eq, respectively, with the ISO 14040/14044 standards, the PAS 2050 and the CEPI framework.

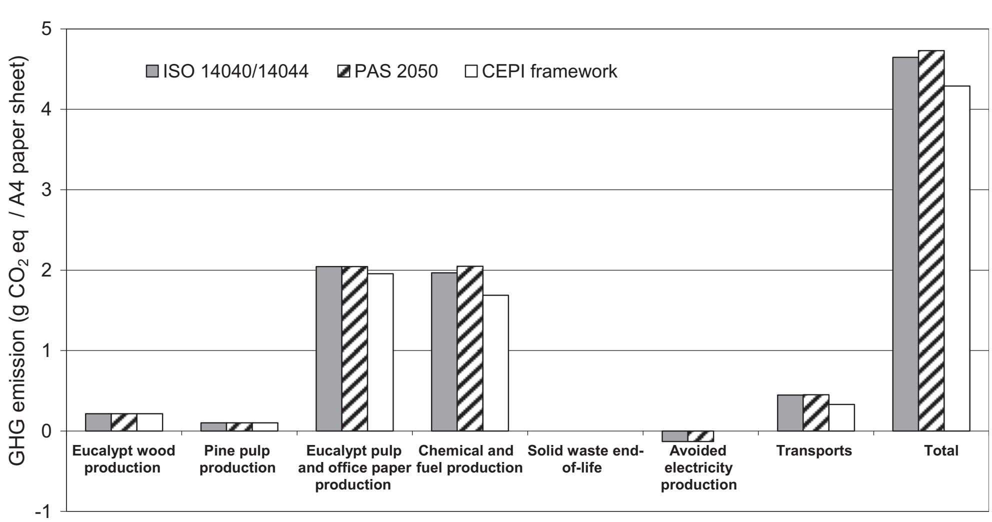

Fig. 2. Carbon footprint of office paper.

These values are lower than the ones estimated for 80 g/m^2 office paper produced from eucalypt wood on the late 1990s (Dias et al., 2002, 2007a; Lopes et al., 2003), which amounted to about 1200 kg CO_2eq per tonne of paper. This is mainly due to the fact that in those studies the fuel used to produce the energy requirements of paper production was fuel oil while in the present study is natural gas, which has smaller GHG emissions per unit of energy produced. It should be noted that in those studies and in the present study the production of the energy used in pulp and paper production processes is one of the main sources of GHG emissions. This was also found in other LCA studies carried out for pulp and paper products (Boguski, 2010; González-García et al., 2009; Nors et al., 2009).

Since the use of different methodologies resulted in different estimations for the carbon footprint of office paper, it is important to further define internationally agreed rules, such as Product Category Rules developed in accordance with ISO 14025 (ISO, 2006c), in order to allow robust comparisons between different paper products or similar paper products with distinct origins or production technologies. The type of GHGs included, the boundaries of the system, the functional unit, the quality of the input data, the allocation procedures applied, between other aspects, should therefore be similar or even the same when determining the carbon footprint.

5. Conclusions

The main conclusions drawn from this study are as follows.

- The use of different methodologies to assess the carbon footprint of office paper gave different results: 4.64, 4.74 and 4.29 g CO_2eq per A4 sheet, respectively, with the ISO 14040/14044 standards, the PAS 2050 and the CEPI framework;
- despite the methodological differences, all methodologies identified the stages of eucalypt pulp and office paper production and chemical and fuel production as the major hot spots;
- the specification of adequate cut-off criteria may decrease the effort needed for data collection without compromising the quality of the results;
- general methodologies such as those analysed in this study are not enough to avoid biased comparisons between products. More specific rules that limit the degree of freedom in the choice of the functional unit, system boundary, allocation rules, data quality, between others, should be developed.

Acknowledgements

Thanks are due to FCT (Science and Technology Foundation – Portugal) for the scholarship granted to Ana Cláudia Dias (SFRH/BPD/1084/2000).

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